

Xuanzhuo Liu

(+86) 180-2507-1157 • xuanzliu@stanford.edu • [personal website](#)

EDUCATION

Stanford University | MS in Electrical Engineering Oct. 2026 – Aug. 2028

Signal Processing, Optimization and Statistical Learning (Advisor: Stephen P. Boyd)

The Chinese University of Hong Kong, Shenzhen Sept. 2022 – May 2026

BS, Data Science and Big Data Technology (Advisor: Prof. Haizhou Li)

- **cGPA:** 3.92 / 4.0 (**Rank:** 1 / 149) **mGPA:** 4.0 / 4.0
- **Honors:** Dean's List, Undergraduate Research Award, Academic Scholarship (Top 1%), Best UG Teaching Fellows

University of Oxford | Visiting Student Oct. 2024 – Aug. 2025

Mathematics and Computer Science (First Class)

- **Key Coursework:** Stochastic Differential Equations, Functional Analysis, Real Analysis, Numerical Analysis, Optimal Control, Bayesian Statistical Probabilistic Programming, Geometric Deep Learning, Graph Representation Learning

TECHNICAL SKILLS

- **C++ (5+ years' experience):** Memory optimization, multi-threaded programming, low-latency inference pipeline
- **Modeling:** Reinforcement learning, sequential decision optimization, transformer architectures, large-scale model training
- **Quantitative:** Statistical modeling, stochastic optimization, time-series analysis, Monte Carlo methods

RESEARCH EXPERIENCE

Microsoft Research Asia | Machine Learning Group [\[Stars of Tomorrow\]](#) July 2025 – Feb. 2026

High-dimensional optimization, low-latency sequential decision systems & on-policy RL

- Built and pretrained a 4B-parameter vision-language-action model from scratch on 10,000+ hours of manipulation data.
- Implemented symmetry-aware encoders to improve sample efficiency by 20% (90+% success with only 100 demonstrations).
- Added a minimum-jerk regularization to reduce action variance by 25%, producing smoother and more stable policy outputs.
- Reduced inference latency by 30% via knowledge distillation while preserving representation quality for real-time control.
- Achieved **Top 2** in 2025 IROS Manipulation Challenge out of 300+ teams, recognized as **Rising Tech Talent**.

Oxford Robotics Institute | Advised by Prof. Shimon Whiteson Oct. 2024 – July 2025

Attention-based sequence models & signal-to-noise analysis

- Developed a PDE-based framework modeling information flow in transformer to diagnose layer-wise information loss.
- Implemented Differential Transformer layers to sharpen critical signals, achieving 80+% success rates in SimplerEnv tasks.
- Conducted controlled ablation studies to quantify the marginal value of multimodal conditioning for generalization.
- Proposed a dual-manifold projection scheme for model-predictive control, decoupling transition dynamics from feasibility constraints, reducing storage cost by 20% while maintaining real-time control at 30 Hz.

Shenzhen Research Institute of Big Data | Advised by Prof. Haizhou Li Sept. 2023 – Aug. 2024

Time-series statistical modeling & sequential decision analysis

- Designed controlled experiments and statistical analysis pipelines for noisy multimodal data in multi-speaker environments.
- Quantified cross-modal attention patterns from eye-tracking signals; co-authored paper accepted at ICSR 2024.
- Used large language models to generate structured diagnostic rule sets for specific diseases from clinical text and guidelines
- Developed a sparse-transformer model with two-stage reinforcement learning to capture doctors' perception-reasoning loop.
- Achieved **~45% accuracy** in predicting diagnostic decisions in MIMIC dataset; work under submission to ICML 2026.

PUBLICATIONS

- [1] Lichuan Jiang, Jiani Zhong, **Xuanzhuo Liu**, Siqi Cai, Haizhou Li, (co-author) "The Impact of Synchronized Visual and Auditory Attention on Human Perception", **Best Student Paper Award**, *the 16th International Conference on Social Robotics*.
- [2] **Xuanzhuo Liu**, Zheng Xiong, Shimon Whiteson, "Differential VLA: Reducing Attention Noise for Robust Vision-Language Action Policy Learning", *submitted 2026 CVPR*.